Game Theory

Complete Course

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Based on Philippe BICH's course
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Acknowledgments
I thank **Prince NIANGA MBOUALA** for sharing his notes, which contributed to the enrichment of this document.

Disclaimer
This document is the fruit of my personal work. It is based on my own notes taken during the Game Theory course, as well as my efforts to understand and structure the taught concepts. It is in no way an official document validated by the teaching team, nor an absolute reference. It should be considered as a study and revision aid, reflecting my own interpretation and learning process.

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Chapter 1

Normal Form Games

1.1 Introduction and Practical Information

- **Professor:** Philippe BICH (bich@univ-paris1.fr)
- **Credits:** 5 ECTS (for Game Theory 1-2)
- **Method:** Lectures and problem-solving sessions.
- **Prerequisites:** Analysis and basic probability theory.
- **References:**
  1. "Game theory" - Maschler, Solan, Zamir
  2. "Bases mathématiques de la théorie des jeux" - Sorin, Laraki, Renault
- **Evaluation:** Attendance, slide exercises (with bonus for starred exercises *), and a final exam featuring similar questions. **Warning:** The quality of argumentation and reasoning counts as much as, if not more than, the calculation result.

1.2 Introductory Concepts

Game theory analyzes situations where the optimal behavior of an individual depends on the choices of others. It is a major interdisciplinary field.

- **Economics:** Apple pricing, auctions.
- **Finance:** Portfolio strategies, pricing of financial instruments.
- **Biology:** Cooperative strategies of species, evolution.
- **Politics:** Nuclear armament, voting, strikes.
- **Networks:** Decision to create a link on a social network, information or advertising transmission.
- **Mathematics & Mean Field Games:** Analysis of situations where behavior depends on the average of a large number of people (e.g., choice of an optimal path in traffic, location choice).

Academic Context: Game theory is a pillar of modern research. **More than a dozen Nobel Prizes** have been awarded to game theorists, many of whom are mathematicians (more than just economists). Notable examples include **John Nash**, popularized by the movie "A Beautiful Mind".

Important Nuances:

- **Schelling's Model (Segregation)**: This model illustrates how weak individual preferences for diversity can lead to total segregation.
  - A grid where each cell is occupied by an agent of type A or B.
  - An agent is satisfied if at least a certain percentage of their neighbors are of their type.
  - Otherwise, they move to the nearest empty cell that satisfies them.

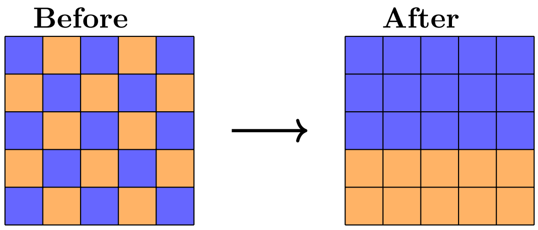


Figure 1.1: Schelling's Model: emergent segregation.

1.3 Normal Form Games

1.3.1 Definitions

Definition 1.1. A **normal form game** is a triplet $(N, (X_i)_{i\in N}, (u_i)_{i\in N})$ where:

- N is the set of players.
- X_i is the set of actions for player $i \in N$.
- $u_i : X \to \mathbb{R}$ is the payoff function.
- $X = \prod_{i\in N} X_i$ represents the set of **action profiles**. A profile is a list $x = (x_1, x_2, \dots, x_n)$ containing an action for each player.

1.3.2 Famous Examples

1. **Coordination Game:**

	A	B
A	(1, 1)	(0, 0)
B	(0, 0)	(2, 2)

2. **Chicken Game / Hawk-Dove:**

	Swerve	Straight
Swerve	(0, 0)	(-1, 1)
Straight	(1, -1)	(-10, -10)

3. **Matching Pennies:**

	Heads	Tails
Heads	(1, -1)	(-1, 1)
Tails	(-1, 1)	(1, -1)

4. **Prisoner's Dilemma:**

	Don't Confess	Confess
Don't Confess	(-1, -1)	(-10, 0)
Confess	(0, -10)	(-5, -5)

5. **Guess two-thirds of the average:** This game illustrates that your optimal strategy depends on your ability to guess the rationality level of other players.

1.3.3 Examples of Continuous Games

First-Price Auction $N = \{1, 2\}$, $X_1 = X_2 = [0, +\infty[$. Values $v_1 = 1M, v_2 = 1.1M$.

$$u_i(x_i, x_j) = \begin{cases} v_i - x_i & \text{if } x_i > x_j \\ \frac{v_i - x_i}{2} & \text{if } x_i = x_j \\ 0 & \text{otherwise} \end{cases}$$

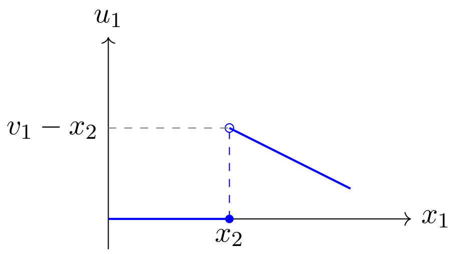


Figure 1.2: Payoff $u_1(x_1)$ for fixed x_2: discontinuity at $x_1 = x_2$.

This graph illustrates the payoff discontinuity in a first-price auction.

- If the player bids $x_1 < x_2$, they lose and get 0.
- If they bid $x_1 = x_2$, they split the payoff 50% (white dot/circle).
- If they bid $x_1 > x_2$ (even by a tiny amount), they win the whole market (abrupt jump in payoff).

This discontinuity prevents the direct application of Nash's theorem (which requires continuous payoff functions).

Second-Price Auction (Vickrey) Two players $i = 1, 2$ with valuations v_i. They submit bids $x_i \geq 0$. The winner pays the loser's bid.

$$u_1(x_1, x_2) = \begin{cases} v_1 - x_2 & \text{if } x_1 > x_2 \\ \frac{v_1 - x_2}{2} & \text{if } x_1 = x_2 \\ 0 & \text{if } x_1 < x_2 \end{cases}$$

1.4 Exercises

Example 1.2 (All pay auction)**.** All bidders pay their bid, whether they win the object or not. The highest bidder wins the object.

- **Mechanism:** Each player submits a bid $x_i \geq 0$.

- **Payoff function:** For player i, with valuation v_i:

$$u_i(x_i, x_{-i}) = \begin{cases} v_i - x_i & \text{if } x_i > x_j \quad \forall j \neq i \\ -x_i & \text{if } x_i < \max(x_{-i}) \end{cases}$$

(In case of a tie for the highest bid, the payoff might be shared).

Example 1.3 (Ice-cream sellers (Hotelling))**. Task:** Two sellers position themselves on the interval $[0, 1]$. Customers are uniformly distributed and go to the nearest seller.

1. Formally write down the payoff functions.
2. Plot $u_1(x_1, x_2)$ for a fixed value $x_2 < 1$.

Solution:

1. Customers choose the nearest seller. The "indifferent" customer is located at the midpoint between the two sellers, i.e., at $m = \frac{x_1+x_2}{2}$. If $x_1 < x_2$, Seller 1 captures all customers in $[0, m]$. Their payoff (market share) is:

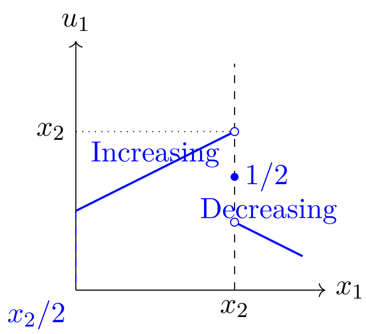
$$u_1(x_1, x_2) = \frac{x_1 + x_2}{2}$$

If $x_1 > x_2$, Seller 1 captures customers in $[m, 1]$. Their payoff is:

$$u_1(x_1, x_2) = 1 - \frac{x_1 + x_2}{2}$$

If $x_1 = x_2$, they share the market: $u_1 = 1/2$.

2. For a fixed x_2 (say $x_2 = 0.7$), here is the shape of $u_1(x_1)$:



The maximum is reached just "before" x_2 (to capture the entire left market). At equilibrium, both position themselves at the center (1/2).

Chapter 2

Dominance

2.1 Important Notation

This notation is fundamental and will be used throughout the course. It may be confusing at first glance.

- $X_{-i} = \prod_{j \in N, j \neq i} X_j$ denotes the set of action profiles of all players **except** player i.
- x_{-i} denotes an element of X_{-i} (a vector of the opponents' actions).
- $u_i(x_i, x_{-i})$ is an abusive notation for $u_i(x)$. It highlights that the payoff depends on the player's choice x_i and the others' choices x_{-i}.

Example 2.1 (3-Player Example)**.** Let $N = \{1, 2, 3\}$. For player $i = 2$:

- The opponents are 1 and 3.
- The opponents' profile is $x_{-2} = (x_1, x_3)$.
- The set of these profiles is $X_{-2} = X_1 \times X_3$.
- The payoff function is written as $u_2(x) = u_2(x_2, x_{-2}) = u_2(x_2, (x_1, x_3))$.

2.2 Dominant Action and Equilibrium

2.2.1 Definitions

Definition 2.2. An action $x_i \in X_i$ is **dominant** if it always yields more (or as much) payoff than any other action, regardless of the others' choices:

$$\forall d_{-i} \in X_{-i}, \forall d_i \in X_i, u_i(x_i, d_{-i}) \geq u_i(d_i, d_{-i})$$

Definition 2.3. A profile $x = (x_i)_{i \in N}$ is a **dominant strategy equilibrium** if for each i, x_i is dominant.

Rationality and Collective Failure: The example below shows that being individually rational (choosing one's dominant strategy) can lead to a collective disaster. This is the central paradox of the Prisoner's Dilemma.

P1 \P2	a	b
A	(1 000 000, 1 000 000)	(0, 1 000 001)
B	(1 000 001, 0)	(100, 100)

Here, playing B is dominant for both, leading to (0.1, 0.1), while cooperation (A, A) yielded 1 Million each.

Remark (Reflection). Can you think of societal examples where playing a dominant action for everyone leads to a bad outcome? (Pollution, arms race, etc.)

2.3 Common Knowledge

Definition 2.4. A statement S is **common knowledge** if:

- Everyone knows that S is true.
- Everyone knows that everyone else knows that S is true.
- Everyone knows that everyone else knows that everyone else knows... ad infinitum.

Nuance: There is a fundamental difference between "I know S" and "S is common knowledge". Your behavior will change radically depending on whether you think the other person is aware or not (examples: coordinated attack, email acknowledgement).

2.4 Iterated Elimination of Strictly Dominated Strategies (IESDS)

2.4.1 Strictly Dominated Action

Definition 2.5. x_i is **strictly dominated** by y_i if:

$$\forall d_{-i} \in X_{-i}, u_i(x_i, d_{-i}) < u_i(y_i, d_{-i})$$

A rational agent will never play a strictly dominated action.

2.4.2 IESDS Process (Formalization)

- We start with the initial game $G = G(0)$ with action sets $X_i(0)$.
- **Step 1:** We eliminate all strictly dominated actions in $G(0)$ to obtain sets $X_i(1)$ and the reduced game $G(1)$.
- **Recurrence:** At step k, we eliminate strictly dominated actions in game $G(k-1)$ to obtain $G(k)$.
- **Result:** The set of surviving profiles is the intersection of all steps:

$$E^\infty = \bigcap_{n \geq 0} X(n)$$

Corollary 2.6 (Nash-IESDS Link)**.** *The set of Nash equilibria is always included in the set of profiles surviving IESDS. If IESDS reduces the game to a single profile, it is the unique Nash equilibrium.*

2.5 Exercises and Methodology

2.5.1 Resolution Method (Exam Type)

To solve a game using IESDS, you must explicitly write down the steps:

1. "In the initial game, for Player i, action A strictly dominates B because $u_i(A, \cdot) > u_i(B, \cdot)$."
2. "We remove action B."
3. "In the reduced game, for Player j, action..."

2.5.2 Application Exercises

Example 2.7 (Application Exercise (Continuous Functions)). Consider a two-player game with payoffs:

$$u_1(x, y) = (-x^2 + x + 1)e^y \quad \text{and} \quad u_2(x, y) = (2y - y^2 + 2)e^x$$

where $x, y \geq 0$.

- For Player 1, maximizing u_1 with respect to x (treating e^y as a positive constant) amounts to maximizing $-x^2 + x + 1$.

$$\frac{\partial u_1}{\partial x} = (-2x + 1)e^y = 0 \implies x = 1/2$$

- For Player 2, maximizing u_2 with respect to y amounts to maximizing $2y - y^2 + 2$.

$$\frac{\partial u_2}{\partial y} = (2 - 2y)e^x = 0 \implies y = 1$$

The unique dominant strategy equilibrium is $(1/2, 1)$.

Example 2.8 (Illustration of IESDS Process). Consider the following game and matrix:

$$\begin{pmatrix} & A & B & C \\ a & (1,-1) & (0,1) & (-1,0) \\ b & (2,3) & (2,2) & (0,1) \\ c & (1,4) & (0,3) & (1,2) \end{pmatrix}$$

Step-by-step resolution:

1. **Step 1:** For Player 1, action a is strictly dominated by b (since $1 < 2, 0 < 2, -1 < 0$). Eliminate row a.
2. **Step 2:** In the reduced game (rows b, c), Player 2 compares their columns.
   - B vs A: $(2 < 3)$ if b, $(3 < 4)$ if c. So B is dominated by A.
   - C vs A: $(1 < 3)$ if b, $(2 < 4)$ if c. So C is dominated by A.

   Eliminate columns B and C.
3. **Step 3:** In the reduced game (column A only), Player 1 chooses between b (payoff 2) and c (payoff 1). Action c is dominated. Eliminate c.
4. **Solution:** The only surviving profile is (b, A) with payoffs $(2, 3)$.

2.6 Relation to Best Response

Proposition 2.9. *A strictly dominated action is* ***never*** *a best response to any belief about opponents' strategies.*

Remark. Fundamental link: This proposition links the behavioral hypothesis of rationality (not playing dominated actions) to the mathematical concept of best response (fixed point, Nash).

Exercise 2.10 (Analysis of Classic Games)**.** • **Prisoner's Dilemma:** Demonstrate that "Confess" is a dominant strategy for each player.

- **Coordination Game and Chicken Game:** Show that no player has a dominant strategy (the best action depends on the other).

Example 2.11 (Exam 2023: IESDS Matrix)**. Question:** Consider a two-player game defined by the following payoff matrix (the first number is Player 1's payoff, the second is Player 2's):

P1 \P2	W	X	Y	Z
A	(1, 0)	(1, 0)	(0, 0)	(-1, 0)
B	(4, 0)	(4, 0)	(1, 2)	(0, 3)
C	(2, 0)	(2, 0)	(0, 2)	(0, 3)
D	(3, 0)	(3, 0)	(3, 2)	(4, 4)

Determine the Nash equilibrium using Iterated Elimination of Strictly Dominated Strategies.
Solution:

1. **Step 1:** For Player 1, action A is strictly dominated by B (since $4 > 1, 4 > 1, 1 > 0, 0 > -1$). Eliminate row A.
2. **Step 2:** For Player 2, in the remaining game, eliminate action X (dominated by other columns).
3. **Step 3:** For Player 1, eliminate action C.
4. **Step 4:** For Player 2, eliminate action W.
5. **Step 5:** For Player 1, eliminate action B.

Result: The profile (D, Z) is the unique equilibrium obtained by elimination. We easily check that (D, Z) is a Nash equilibrium.

Example 2.12 (Theoretical Proof)**.** Let G be a game where u_i are continuous and X_i are non-empty compact sets. **Question:** Prove that the set of surviving equilibria under IESDS is compact and non-empty.

Solution: Let $X^0 = \prod X_i$ be non-empty compact. At each step k, we remove strictly dominated strategies. If u_i is continuous, the set of dominated strategies is open (or a union of open sets), so the remaining set X^k is closed. Since $X^k \subset X^{k-1} \subset \cdots \subset X^0$, X^k is a decreasing sequence of compact sets. Assuming the elimination is well-defined (never empty at each finite step thanks to the maximum theorem), the infinite intersection $X^\infty = \bigcap_{k \geq 0} X^k$ is a decreasing intersection of non-empty compact sets in a Hausdorff space. By the Borel-Lebesgue property (or Cantor's intersection theorem), this intersection is **non-empty and compact**.

Example 2.13 (Classic Check)**.** Are there dominant strategies in:

- The Coordination Game? **No.** (Depends on the other).
- The Chicken Game? **No.** (If the other swerves, I swerve? No. If they swerve, I go straight; if they go straight, I swerve).
- The Prisoner's Dilemma? **Yes.** "Confess" is a strictly dominant strategy for both players,

regardless of the other's action.

Chapter 3

Nash Equilibrium (Pure Strategies)

This chapter introduces the most famous and widely used prediction notion in game theory: the Nash equilibrium.

3.1 Introductory Example and Intuition

Consider a 2-player game with $X_1 = X_2 = [0, 1]$. Payoff functions are quadratic (concave):

- $u_1(x_1, x_2) = -(x_1 - x_2)^2$
- $u_2(x_1, x_2) = -(1 - x_1 - x_2)^2$

Analysis:

1. For player 1, maximizing u_1 means minimizing the distance between x_1 and x_2. Their best response is therefore $x_1 = x_2$.
2. For player 2, maximizing u_2 means choosing $x_2 = 1 - x_1$.
3. The equilibrium is reached when both strategies are mutually optimal: $x_1 = x_2 = \frac{1}{2}$.

Remark (Fundamental Contrast: Nash vs Dominance). The profile $(\frac{1}{2}, \frac{1}{2})$ is a Nash equilibrium, but it is **NOT** a dominant strategy equilibrium.

Proof. For player 1, action $\frac{1}{2}$ is not dominant. Indeed:

- If P2 plays $x_2 = \frac{1}{2}$, then $u_1(\frac{1}{2}, \frac{1}{2}) = 0$ (maximum possible).
- But if P2 plays $x_2 = 0$, then $u_1(\frac{1}{2}, 0) = -(\frac{1}{2})^2 = -0.25$.
- However, if P1 had played 0 against 0, they would have had $u_1(0, 0) = 0$.
- Thus, playing $\frac{1}{2}$ is not always better than playing 0.

This illustrates that Nash equilibrium is a weaker stability condition than dominance. □

3.2 Formal Definitions

Definition 3.1 (Nash Equilibrium)**.** Consider a normal form game $(N, (X_i)_{i\in N}, (u_i)_{i\in N})$. An action profile $\bar{x} = (\bar{x}_i)_{i\in N}$ is a **Nash equilibrium** if for each player i, no unilateral deviation is profitable:

$$\forall i \in N, \forall d_i \in X_i, \quad u_i(\bar{x}_i, \bar{x}_{-i}) \geq u_i(d_i, \bar{x}_{-i})$$

3.3 Best-reply Correspondence

Definition 3.2 (Best-reply Correspondence)**.** The best response of player i is not necessarily a unique function; it is a **correspondence** (a set-valued function), denoted $B_i : X_{-i} \rightrightarrows X_i$:

$$B_i(x_{-i}) := \{x_i \in X_i : \forall d_i \in X_i, u_i(x_i, x_{-i}) \geq u_i(d_i, x_{-i})\}$$

Definition 3.3 (Global Correspondence)**.** We define $B : X \rightrightarrows X$ as the Cartesian product of individual best responses:

$$B(x) = \prod_{i \in N} B_i(x_{-i})$$

Theorem 3.4 (Fixed Point Characterization)**.** *An action profile $x \in X$ is a Nash equilibrium if and only if:*

1. *x is a **fixed point** of the correspondence B, i.e., $x \in B(x)$.*
2. *Geometrically, x belongs to the **intersection of the graphs** of the players' best responses.*

This link with fixed points is crucial because it will allow the use of powerful topological theorems (Brouwer, Kakutani) to prove the existence of equilibria.

3.4 Graphical Examples: BR Intersection

For two-player games with continuous actions (e.g., $X_1 = X_2 = [0, 1]$), Nash equilibria correspond to the intersection points of the curves $x_1 = B_1(x_2)$ and $x_2 = B_2(x_1)$ in the (x_1, x_2) plane.

- **Unique Equilibrium:** The curves intersect at a single point (classic case of concave functions like the introductory example).
- **Multiple Equilibria:** The curves intersect at several points. For example, in a coordination game (continuous Battle of Sexes), there can be two stable intersections and one unstable one.
- **No Equilibrium (in pure strategies):** The curves never intersect. This is the case for the continuous "Matching Pennies" game without a fixed point.

3.5 Relation to Dominance

Corollary 3.5 (Inclusion in IESDS)**.** *The set of Nash equilibria is included in the set of action profiles that survive Iterated Elimination of Strictly Dominated Strategies (IESDS).*

Remark. This is a powerful result: IESDS, often simpler to compute, can be used as a "bounding box" to reduce the search space for Nash equilibria.

Proposition 3.6 (Exclusion Property)**.** *If an action x_i is strictly dominated, then it is **never** a best response (it never belongs to $B_i(x_{-i})$ for any x_{-i}).*

Note: The converse is not always true (except in finite 2-player zero-sum games, for example).

Proposition 3.7 (Stability by Elimination)**.** *Let G' be the game obtained by removing a strictly dominated action of a player. Then the set of Nash equilibria of G is identical to that of G'.*

This property justifies the use of IESDS as a preliminary step to simplify a game before searching for its Nash equilibria.

3.5.1 Rigorous Resolution Example (IESDS)

P1 \P2	A	B	C
a	(2,2)	(1,1)	(1,0)
b	(0,2)	(0,3)	(1,1)
c	(0,1)	(-1,-1)	(2,0)

Expected Write-up:

1. **Step 1:** For P2, action A strictly dominates C (since $2 > 0, 3 > 1, 1 > 0$). Eliminate column C.
2. **Step 2:** In the reduced game, for P1, action a strictly dominates b (since $2 > 0, 1 > 0$) and c (since $2 > 0, 1 > -1$). Eliminate rows b and c.
3. **Step 3:** Only row a remains. For P2, facing a, A yields 2 and B yields 1. A strictly dominates B.

Conclusion: The unique residual equilibrium is (a, A). It is the unique Nash equilibrium.

3.6 Exercises and Resolution Methods

3.6.1 Method for Continuous Games

If payoff functions are differentiable and concave with respect to the player's variable:

1. Compute the First Order Condition (FOC): $\frac{\partial u_i}{\partial x_i} = 0$.
2. Isolate x_i to find the best response function $x_i = BR_i(x_{-i})$.
3. Solve the system of equations formed by all best responses to find the fixed point.

3.6.2 Application Exercises

Example 3.8 (Exam 2023)**.**
- A) $u_1(x, y) = xy - x^2 + 2$, $u_2(x, y) = 2xy - y^2$ on $[0, 1]^2$. Calculate equilibria using the FOC method.
- B) $u_i(x_1, x_2) = x_i$ if $x_1 + x_2 \leq 1$ and 0 otherwise. Calculate the Nash set N and Pareto optimal points.

Solution:

- **A)** FOC Player 1: $\frac{\partial u_1}{\partial x} = y - 2x = 0 \implies x = y/2$. FOC Player 2: $\frac{\partial u_2}{\partial y} = 2x - 2y = 0 \implies y = x$. System: $x = x/2 \implies x = 0, y = 0$. Unique equilibrium $(0, 0)$. Hessians: $\partial^2 u_1/\partial x^2 = -2 < 0$ (concave), same for P2. It is indeed a maximum.
- **B)** - If $x_1 + x_2 < 1$, each player wants to increase their x_i (increasing payoff) until the constraint acts. Not stable. - If $x_1 + x_2 > 1$, payoff 0. They have an incentive to reduce to get under 1. - Equilibrium: Any pair (x_1, x_2) such that $x_1 + x_2 = 1$ and $x_i \geq 0$. If a player deviates upwards, payoff 0. If downwards, reduced payoff. - Pareto: Any point on the frontier $x_1 + x_2 = 1$ is Pareto-optimal (sum 1).

Example 3.9 (Exam 2023: Nash and Best Responses)**. Question:** Consider a continuous game. Determine Player 1's best response to y and the Nash equilibria of the game for various parameters.
Solution:

- Player 1's best response to y is the solution to $\sup_{x \in \mathbb{R}} u_1(x, y)$.
- **Case 1:** If there is no solution because $\lim_{x \to \infty} u_1(x, y) = \infty$, then there is no Nash equilibrium.
- **Case 2:** If the function is concave, the solution satisfies the necessary first-order condition (FOC). For example, for Player 2, y must satisfy $4y = 7/2$ in a given configuration.
- The best response correspondences (BR_1 and BR_2) are analyzed on $[0, 1]$. The Nash equilibrium is found at the intersection of these correspondences.

Example 3.10 (Rent-seeking / Tullock)**.** n lobbyists choose effort $x_i \geq 0$. Cost cx_i. Gain if win: V. Payoff: $\frac{x_i}{\sum x_j} V - cx_i$. **Hint:** Look for a **symmetric equilibrium** where $x_i = x^*$ for all i, which greatly simplifies solving the FOC.

Solution: Each player i maximizes $u_i(x_1, \ldots, x_n) = \frac{x_i}{x_i + \sum_{j \neq i} x_j} V - cx_i$. FOC: $\frac{\partial u_i}{\partial x_i} = \frac{1 \cdot (\sum x_j) - x_i \cdot 0}{(\sum x_j)^2} V - c = 0$? No, quotient derivative. $\frac{\partial u_i}{\partial x_i} = V \frac{1 \cdot (x_i + X_{-i}) - x_i \cdot 1}{(x_i + X_{-i})^2} - c = V \frac{X_{-i}}{(\sum x_j)^2} - c = 0$. At symmetric equilibrium $x_i = x^*$. $\sum x_j = nx^*$. $X_{-i} = (n-1)x^*$. $V \frac{(n-1)x^*}{(nx^*)^2} = c \implies V \frac{n-1}{n^2 x^*} = c \implies x^* = \frac{n-1}{n^2} \frac{V}{c}$.

Example 3.11 (Exam 2023: Symmetric Nash Equilibrium and Optimality)**. Question:** Consider a symmetric game. Find the symmetric Nash equilibrium and determine if it is socially optimal.
Solution:

- In a symmetric equilibrium, each Player i maximizes their payoff u_i. We look for p such that the derivative of the payoff function is zero ($f'(p) = 0$).
- In this exercise, we find a symmetric equilibrium $\bar{p} = 1/4$ (or values depending on n).
- We then compare the equilibrium payoff with the social maximum (Max $\sum u_i$).
- **Conclusion:** The Nash equilibrium here is suboptimal from a social point of view.

Example 3.12 (Public Good)**.** n inhabitants. Wealth W. Contribution x_i. Payoff: $u_i(x) = W - x_i + \sqrt{\sum_{j=1}^{n} x_j}$. Show that the individual contribution at equilibrium decreases with n.

Solution: We look for a symmetric equilibrium $x_i = x^*$. First order condition (player i considers others' x_j as fixed, let $S_{-i} = \sum_{j \neq i} x_j$):

$$\frac{\partial u_i}{\partial x_i} = -1 + \frac{1}{2\sqrt{x_i + S_{-i}}} = 0$$

At symmetric equilibrium, $S_{-i} = (n-1)x^*$ and $x_i = x^*$.

$$1 = \frac{1}{2\sqrt{nx^*}} \implies 2\sqrt{nx^*} = 1 \implies 4nx^* = 1 \implies x^* = \frac{1}{4n}$$

The individual contribution $x^*(n) = \frac{1}{4n}$ is indeed a decreasing function of n (free-rider phenomenon).

Chapter 4

Existence of Nash Equilibrium

4.1 Non-existence of Nash Equilibrium

The existence of a Nash equilibrium is not guaranteed. Several obstacles can prevent it.

4.1.1 Examples of Non-existence

1. **Lack of compactness:** 2 players on $[0, +\infty[$, $u_1 = x_1$, $u_2 = x_2$. No equilibrium because payoffs are unbounded (everyone wants to play $+\infty$).

2. **Lack of convexity or continuity (Continuous Matching Pennies):** This is the game "I like you, you don't like me". $u_1(x_1, x_2) = -|x_1 - x_2|$ (wants to get closer) and $u_2(x_1, x_2) = |x_1 - x_2|$ (wants to move away) on $[0, 1]$.

   - If $x_1 = x_2$, P2 wants to move away (play 0 or 1).
   - If $x_1 \neq x_2$, P1 wants to get closer to x_2.
   - There is no fixed point, so no pure strategy equilibrium.

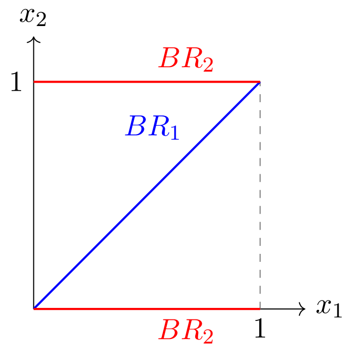


Figure 4.1: Continuous Matching Pennies: best responses never intersect.

> In this continuous zero-sum game on the square (Matching Pennies), the best responses (blue and red lines) never cross. There is therefore no intersection of the best response graphs, which implies that no pure strategy Nash equilibrium exists.

Remark (Cultural Illustration). The Professor often associates this frustration of non-existence with the painting **"The Scream" by Edvard Munch**: the anxiety of not finding a stable solution in a world without convexity or compactness.

4.1.2 Obstacles to Existence

The main obstacles to the existence of a Nash equilibrium are:

- **Non-convex action sets:** In finite games, strategy sets have "holes".
- **Unbounded or non-closed action sets:** Compactness problem.
- **Discontinuous payoff functions.**
- **Lack of quasi-concavity:** If the payoff function is not quasi-concave, the set of best responses may not be convex.

4.2 Quasiconcavity and Risk Appetite

Quasi-concavity is a crucial property for existence. It is intimately linked to the notion of risk aversion or risk-seeking.

Definition 4.1 (Quasiconcavity)**.** Let X be a convex set. $f : X \to \mathbb{R}$ is **quasiconcave** if for all $\lambda \in [0,1]$ and all $(x, y) \in X^2$:

$$f(\lambda x + (1-\lambda)y) \geq \min\{f(x), f(y)\}$$

Equivalent characterization: f is quasiconcave if and only if all its **upper level sets** are convex:

$$\forall \alpha \in \mathbb{R}, \quad S_\alpha(f) = \{x \in X : f(x) \geq \alpha\} \text{ is convex.}$$

Example 4.2 (Economic Intuition: Risk-Loving)**.** Consider $u(x) = x^2$ on $\mathbb{R}$.

- This function is convex (parabola).
- It is **not quasiconcave** on $\mathbb{R}$ (e.g., $S_1(f) =]-\infty, -1] \cup [1, +\infty[$ is not convex).
- **Interpretation:** An agent with such utility is "Risk-Loving". They prefer an uncertain outcome (e.g., a lottery between -2 and 2, mean 0) to its certain equivalent (0), because $u(0) = 0 < 0.5u(-2) + 0.5u(2) = 4$.

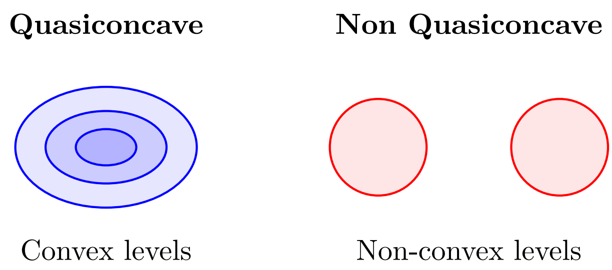


Figure 4.2: Upper level sets: convex (quasiconcave) vs non-convex.

Quasi-concavity is characterized by the convexity of upper level sets $S_\alpha = \{x \mid f(x) \geq \alpha\}$.

- Left: S_α (colored zones) is always convex. The function is quasi-concave.
- Right: For some α, S_α is disjoint (union of two separate sets), so not convex. The function is not quasi-concave.

4.3 Glicksberg's Theorem

This is the fundamental theorem ensuring the existence of a Nash equilibrium in pure strategies.

Theorem 4.3 (Glicksberg)**.** *Let a game $G = (N, (X_i)_{i\in N}, (u_i)_{i\in N})$ such that:*

1. *X_i is a **compact and convex** subset of a topological vector space.*
2. *u_i is **continuous** on $X = \prod X_i$.*
3. *$x_i \mapsto u_i(x_i, x_{-i})$ is **quasiconcave** for all x_{-i}.*

Then, there exists a Nash equilibrium.

4.3.1 Proof (Exam Rigor)

The proof relies on applying the fixed point theorem to the best response correspondence $B(x) = \prod B_i(x_{-i})$.

Why Kakutani and not Brouwer? Brouwer's Theorem applies to continuous **functions**. Here, the best response is not unique (it's a set), it is a **correspondence**. We must therefore use **Kakutani's Theorem**.

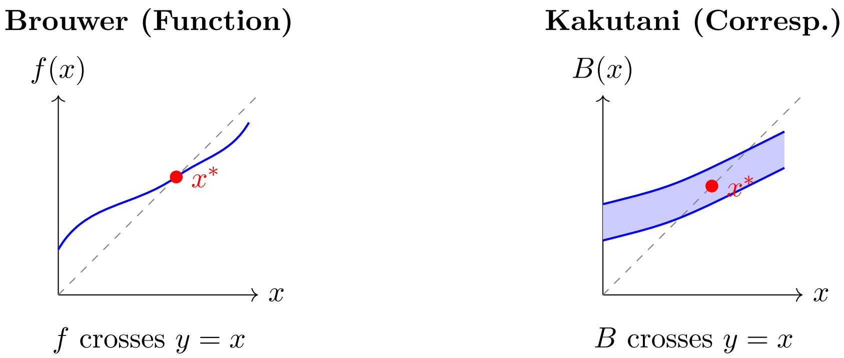


Figure 4.3: Brouwer vs Kakutani: function vs correspondence.

Comparison of theorems:

- **Brouwer (Left):** If f is a continuous *function* from $[0, 1]$ to itself, its graph (blue curve) must necessarily cross the diagonal (dashed). The intersection is a fixed point $x^* = f(x^*)$.
- **Kakutani (Right):** If B is a *correspondence* (set of values) with closed graph and convex values (the blue zone is "solid" vertically), its graph must also cross the diagonal. The intersection x^* satisfies $x^* \in B(x^*)$, it is a fixed point.

Verification of Kakutani's 3 conditions: To apply Kakutani to $B : X \rightrightarrows X$, we must verify:

1. **Non-empty:** Since u_i is continuous on a compact set X_i, it reaches its maximum (Weierstrass Theorem). So $B_i(x_{-i}) \neq \emptyset$.
2. **Convex values:** Since u_i is quasi-concave with respect to x_i, the set of maximizers $B_i(x_{-i})$ is convex.

3. **Closed Graph:** This is a consequence of the global continuity of u_i (Berge's Maximum Theorem). If a sequence $x^n \to x$ and $y^n \in B(x^n)$ with $y^n \to y$, then $y \in B(x)$.

All conditions being met, B admits a fixed point $x^* \in B(x^*)$, which is a Nash equilibrium.

4.4 Solved Examples

4.4.1 Public Good Provision

For n inhabitants, $u_i = W - x_i + \sqrt{\sum x_j}$. We look for a symmetric equilibrium $(a, \dots, a)$. Let $f(x_i) = W - x_i + \sqrt{(n-1)a + x_i}$. The first order condition $f'(a) = 0$ gives:

$$-1 + \frac{1}{2\sqrt{na}} = 0 \iff 2\sqrt{na} = 1 \iff 4na = 1 \iff a = \frac{1}{4n}$$

4.4.2 Rent-seeking (Tullock)

n firms, payoff $u_i = \frac{x_i}{\sum x_j} V - cx_i$. For a symmetric equilibrium a:

$$f'(x_i) = \frac{V((n-1)a + x_i) - Vx_i}{((n-1)a + x_i)^2} - c = 0$$

At $x_i = a$, we get $\frac{(n-1)V}{n^2 a} = c$, so $a = \frac{n-1}{cn^2} V$.

4.5 Extensions and Limiting Cases (Further Reading)

- **Coercivity Argument:** If action sets are not compact (e.g., $X_i = \mathbb{R}$), an equilibrium can still exist if functions are **coercive**, i.e., if $\lim_{|x_i| \to \infty} u_i(x) = -\infty$. This allows restricting analysis to a sufficiently large compact subset.

Example 4.4 (Exam 2023: Proof of Coercivity)**. Question:** Prove the existence of a Nash equilibrium for a game where utility functions are coercive.
Solution:

1. By hypothesis, $\lim_{|x| \to \infty} u_i(x, y) = -\infty$. The function is therefore coercive on a closed set, which guarantees the existence of a maximum and thus a best response.

2. By considering a sequence of games on compact sets $X_n \times Y_n$, we apply Glicksberg's theorem to guarantee the existence of a Nash equilibrium $(\bar{x}_n, \bar{y}_n)$.

3. Since the best responses are bounded by coercivity, the sequence converges to an equilibrium $(\bar{x}, \bar{y})$ of the original game.

4.5.1 Remarks on Existence (Beyond Exam Scope)

In many economic models, payoff functions are discontinuous. The existence of a Nash equilibrium can still be guaranteed under certain conditions:

- **Reny (1999)**: Concept of "better-reply security".
- **Bich-Laraki (2017)**: Work on games with discontinuous payoffs.
- **Bich (2004)**: Non-continuous version of Brouwer's fixed point theorem.

Consensus Game Three players $i \in \{1, 2, 3\}$, $X_i = [0, 1]$.

$$u_1(x_1, x_2, x_3) = -\left|x_1 - \frac{x_2 + x_3}{2}\right|^2$$

Each player seeks to be as close as possible to the average of the others. The equilibrium is $(1/2, 1/2, 1/2)$ if looking for a fixed point.

Exercise 4.5 (Symmetric Nash Equilibrium)**.** Let $G = (X, X, u, u)$ be a symmetric game where X is compact convex and u is continuous and quasi-concave with respect to its first variable. Prove using Kakutani's theorem applied to the best response correspondence BR that there exists a symmetric equilibrium (x^*, x^*).

Solution: 1. Let $BR : X \rightrightarrows X$ be the best response correspondence of the representative player. 2. Define the function $\phi : X \to \mathcal{P}(X)$ by $\phi(x) = BR(x, x, \ldots, x)$? No, careful with arguments.

Simpler method: Let the "symmetric best-reply reaction" correspondence $F : X \rightrightarrows X$ be defined by:

$$y \in F(x) \iff y \in \arg\max_{z \in X} u(z, x, \ldots, x)$$

(y is the best response of a player if ALL others play x).

- X is non-empty compact convex.
- $F(x)$ is non-empty (continuity on compact) and convex (quasi-concavity).
- F has a closed graph (Maximum Theorem).

By Kakutani's Theorem, F admits a fixed point $x^* \in F(x^*)$. This means that if all others play x^*, the best response for me is also to play x^*. Since the game is symmetric, $(x^*, \ldots, x^*)$ is a Nash equilibrium.

4.6 Conclusion and Transition

We have seen that the existence of a Nash equilibrium in pure strategies requires strong conditions (convexity, continuity, quasi-concavity). When these conditions are not met (for example, in finite games where discrete action sets are not convex), a pure strategy equilibrium may not exist (e.g., Matching Pennies, Rock-Paper-Scissors).

To guarantee the existence of a stable solution in all finite games, we must extend our concept of strategy and allow players to randomize their choices. This is the subject of the next chapter on **mixed strategies**.

Chapter 5

Mixed Strategies

5.1 Introduction and Motivation

In some games (e.g., striker vs. goalkeeper, Rock-Paper-Scissors), a pure strategy equilibrium does not exist. If the striker knows the goalkeeper's deterministic action, they can always win. The real equilibrium requires randomness, or more precisely, unpredictability.

5.2 Formal Definitions

5.2.1 Mixed Strategy and Representations

Definition 5.1 (Mixed Strategy)**.** A mixed strategy of player i is a probability measure on their action set X_i.

- If X_i is finite, it is a vector $\sigma_i \in \Delta(X_i) = \{p \in [0,1]^{|X_i|} : \sum p_k = 1\}$.
- If X_i is continuous (compact), it is a regular probability measure μ_i on the Borel sigma-algebra of X_i.

Definition 5.2 (Specific Vocabulary)**.**
- **Pure Strategy (Dirac):** A pure strategy $a_i \in X_i$ can be seen as a degenerate mixed strategy that puts all the weight on a_i. It is denoted δ_{a_i} (Dirac measure).
- **Totally Mixed Strategy:** A mixed strategy σ_i is said to be **totally mixed** if its support is complete, i.e., if it assigns a strictly positive probability to each possible pure action: $\forall x_i \in X_i, \sigma_i(x_i) > 0$.

Triple Representation: A mixed strategy can be viewed in three complementary ways:

1. **Geometric:** A point in the simplex $\Delta(X_i)$.
2. **Functional:** A mass function (discrete case) or density function.
3. **Measure:** A probability measure μ on the set of Borel sets.

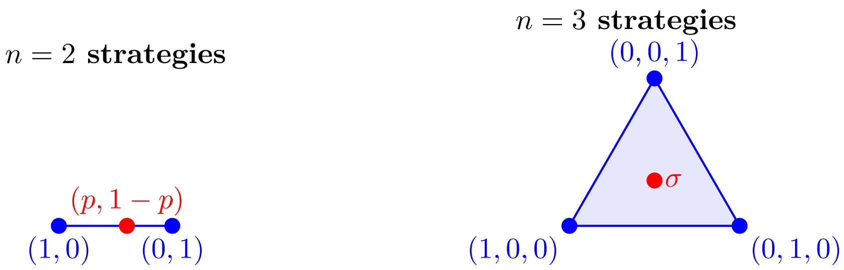


Figure 5.1: The simplex $\Delta(X_i)$: segment for 2 strategies, triangle for 3.

5.2.2 Mixed Extension

The mixed extension is the "new" game where players choose probability distributions.

Definition 5.3 (Expected Payoff and Independence Assumption)**.** Player i's payoff in the mixed extension is defined as the **expected payoff** with respect to the product law of individual strategies. **Fundamental Assumption:** Players' random choices are assumed to be **independent**. The mixed strategy profile is therefore the product measure $P = \sigma_1 \otimes \sigma_2 \otimes \cdots \otimes \sigma_n$ on the space $X = \prod X_i$.
Integral Formulation:

$$\tilde{u}_i(\sigma_1, \ldots, \sigma_n) = \int_{X_1} \cdots \int_{X_n} u_i(x_1, \ldots, x_n) d\sigma_1(x_1) \ldots d\sigma_n(x_n)$$

5.2.3 Mixed strategies in continuous universe

If the action set X_i is a metric space (e.g., $[0, 1]$), a mixed strategy σ_i is a **Borel probability measure** on X_i. We denote $\Delta(X_i)$ the set of these measures. The expected payoff is then:

$$u_i(\sigma_1, \ldots, \sigma_n) = \int_{X_1} \cdots \int_{X_n} u_i(x_1, \ldots, x_n) d\sigma_1(x_1) \ldots d\sigma_n(x_n)$$

In the finite case:

$$\tilde{u}_i(\sigma) = \sum_{x \in X} u_i(x) \left(\prod_{j \in N} \sigma_j(x_j) \right)$$

5.3 Nash Equilibrium in Mixed Strategies

5.3.1 Existence and Link to Pure Strategies

Theorem 5.4 (Nash, 1950)**.** *Every finite game has at least one Nash equilibrium in mixed strategies.*

Proposition 5.5 (Pure/Mixed Relation)**.** *The mixed extension is a coherent generalization. If $x^* = (x_1^*, \ldots, x_n^*)$ is a pure strategy Nash equilibrium of the original game, then the profile of Dirac measures $(\delta_{x_1^*}, \ldots, \delta_{x_n^*})$ is a Nash equilibrium of the mixed extension.*

5.3.2 Indifference Principle (Fundamental)

This is the main tool for calculating mixed equilibria.

Proposition 5.6 (Indifference Principle). *Let σ^* be a mixed Nash equilibrium. For any player i, any pure action x_i belonging to the **support** of their strategy (i.e., played with positive probability $\sigma_i^*(x_i) > 0$) must yield the same expected payoff:*

$$u_i(x_i, \sigma_{-i}^*) = \max_{a_i \in X_i} u_i(a_i, \sigma_{-i}^*) = \tilde{u}_i(\sigma^*)$$

*In other words, the player is **indifferent** between all the actions they actually play. Unplayed actions yield a lower or equal payoff.*

Proof by contradiction. Suppose there exists an action a in the support of σ_i^* such that $u_i(a, \sigma_{-i}^*) < M$, where $M = \max_z u_i(z, \sigma_{-i}^*)$.

1. Let b be an action that achieves the maximum M.
2. Construct a new strategy σ_i' by transferring weight (probability) $p_a > 0$ from the "mediocre" action a to the "optimal" action b.
3. The expected payoff becomes:
$$\begin{aligned}\tilde{u}_i(\sigma_i', \sigma_{-i}^*) &= \tilde{u}_i(\sigma_i^*, \sigma_{-i}^*) - p_a \underbrace{u_i(a, \sigma_{-i}^*)}_{<M} + p_a \underbrace{u_i(b, \sigma_{-i}^*)}_{M} \\ &> \tilde{u}_i(\sigma_i^*, \sigma_{-i}^*)\end{aligned}$$
4. Since $u_i(b) > u_i(a)$, we have $\tilde{u}_i(\sigma') > \tilde{u}_i(\sigma^*)$.

This contradicts the fact that σ^* was a best response (Nash equilibrium). Therefore, all actions in the support must yield the maximum M. □

5.4 Topological and Measure Foundations

This section details the rigorous mathematical framework necessary for infinite games.

- **Borel Sigma-Algebra:** Each action space X_i (assumed compact metric) is equipped with its Borel sigma-algebra $\mathcal{B}(X_i)$, the smallest σ-algebra containing all open sets.
- **Regular Measures:** We consider the space $\mathcal{M}(X_i)$ of regular probability measures (any measure on a compact metric space is regular, i.e., internally approximable by compacts and externally by open sets).
- **Weak-* Topology:** The convergence of a sequence of strategies $\mu_n \to \mu$ is defined by the convergence of integrals against any continuous test function f:
$$\forall f \in C^0(X_i, \mathbb{R}), \quad \int_{X_i} f d\mu_n \to \int_{X_i} f d\mu$$
- **Semi-norms:** This topology is induced by the family of semi-norms $P_f(\mu) = |\int f d\mu|$.

Theorem 5.7 (Weak Compactness). *If X_i is compact metric, then the set of mixed strategies $\Delta(X_i)$ is compact for the weak-* topology.*

It is this theorem, combined with the linearity (thus quasi-concavity) of expected payoffs, that allows applying Glicksberg's theorem to prove the existence of equilibria in continuous games.

5.5 Classic Examples

Example 5.8 (Detailed 2x2 Calculation Example)**.** Consider the following game, where Player 1 (rows) chooses between X and Y, and Player 2 (columns) between x and y.

	x (q)	y ($1-q$)
X (p)	(2, 1)	(0, 0)
Y ($1-p$)	(0, 0)	(1, 2)

Player 1 plays X with probability p, Player 2 plays x with probability q.
Calculation for Player 1 (Indifference): For P1 to be willing to mix (play randomly), they must be indifferent between X and Y. Their expected payoffs must be equal:

$$E[U_1(X, \sigma_2)] = E[U_1(Y, \sigma_2)]$$

$$q \cdot 2 + (1-q) \cdot 0 = q \cdot 0 + (1-q) \cdot 1$$

$$2q = 1 - q \implies 3q = 1 \implies q = 1/3$$

Calculation for Player 2 (Indifference): Similarly, P2 must be indifferent between x and y:

$$E[U_2(\sigma_1, x)] = E[U_2(\sigma_1, y)]$$

$$p \cdot 1 + (1-p) \cdot 0 = p \cdot 0 + (1-p) \cdot 2$$

$$p = 2(1-p) \implies p = 2 - 2p \implies 3p = 2 \implies p = 2/3$$

Conclusion: The unique mixed strategy Nash equilibrium is the profile $((p = 2/3), (q = 1/3))$. Graphically, the best response correspondences intersect at this point $(2/3, 1/3)$ in the probability square.

Example 5.9 (Chicken Game)**.** P1 Indifference: $-4q + (1-q) = -q \implies q = 1/4$.

Chapter 6

Zero-Sum Games

6.1 Topological Preliminaries

Definition 6.1 (Semi-continuity)**.** A function $f : X \to \mathbb{R}$ is **upper semi-continuous (u.s.c.)** if for all $\alpha \in \mathbb{R}$, the set $\{x \in X : f(x) \geq \alpha\}$ is closed.
A function $f : X \to \mathbb{R}$ is **lower semi-continuous (l.s.c.)** if for all $\alpha \in \mathbb{R}$, the set $\{x \in X : f(x) \leq \alpha\}$ is closed.

Exercise 6.2 (Properties to Prove)**.** • Show that the infimum of a family of u.s.c. functions is u.s.c.

- Show that the supremum of a family of l.s.c. functions is l.s.c.

Solution:

1. Let $f = \inf_{i \in I} f_i$ where each f_i is u.s.c. $\{x : f(x) \geq \alpha\} = \{x : \forall i, f_i(x) \geq \alpha\} = \bigcap_{i \in I}\{x : f_i(x) \geq \alpha\}$. Each set $\{f_i \geq \alpha\}$ is closed (def u.s.c.). Any intersection of closed sets is closed. Thus f is u.s.c.

2. Let $g = \sup f_i$ with f_i l.s.c. $\{x : g(x) \leq \alpha\} = \{x : \sup f_i(x) \leq \alpha\} = \{x : \forall i, f_i(x) \leq \alpha\} = \bigcap\{f_i \leq \alpha\}$. Intersection of closed sets $\implies$ closed. Thus g is l.s.c.

Exercise 6.3 (Characterization of quasi-concavity)**.** Prove that a function $f : X \to \mathbb{R}$ is quasi-concave if and only if, for all $\alpha \in \mathbb{R}$, the upper level set $L_\alpha = \{x \in X : f(x) \geq \alpha\}$ is a convex set.

Solution:

- $\Rightarrow$: Assume f is quasi-concave. By definition $f(\lambda x + (1-\lambda)y) \geq \min(f(x), f(y))$. Let $x, y \in L_\alpha$. Then $f(x) \geq \alpha$ and $f(y) \geq \alpha$. So $\min \geq \alpha$. Hence $f(z) \geq \alpha$ for all z in segment $[x, y]$. So $z \in L_\alpha$. L_α is convex.

- $\Leftarrow$: Assume L_α are convex. Let $\alpha = \min(f(x), f(y))$. Then $x \in L_\alpha$ and $y \in L_\alpha$. By convexity, the connecting segment is in L_α, so $f(\lambda x + \dots) \geq \alpha$, Q.E.D.

6.2 Definitions and Analogies

Definition 6.4 (Two-Player Zero-Sum Game)**.** A zero-sum game is a triplet (X, Y, u) where:

- X is the set of strategies for Player 1 (maximizing).
- Y is the set of strategies for Player 2 (minimizing).
- $u : X \times Y \to \mathbb{R}$ represents Player 1's payoff.
- Player 2's payoff is **identically opposite**: $-u(x, y)$.

This model captures the essence of **pure conflict**: what one gains, the other necessarily loses.

Sports and Classical Analogies:

- **Penalty Kick in Football:** If the striker scores (+1), the goalkeeper concedes (-1). It is the archetype of the zero-sum game.
- **Matching Pennies:** Each player chooses H or T. If choices match, P1 wins 1 (P2 loses 1). Otherwise, P1 loses 1 (P2 wins 1).

$$M = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

6.3 Maximin and Minimax Values (Intuition)

Let $v_1(x) = \inf_{y \in Y} u(x, y)$ be the worst possible outcome for P1 if they play x. Player 1, being prudent, will seek to maximize this safety:

$$\underline{v} = \sup_{x \in X} \inf_{y \in Y} u(x, y) \quad \text{(Maximin Value or Security Level)}$$

Similarly, Player 2 wants to minimize P1's payoff (thus maximize their own). They anticipate P1's best move:

$$\overline{v} = \inf_{y \in Y} \sup_{x \in X} u(x, y) \quad \text{(Minimax Value or Ceiling Payoff)}$$

Definition 6.5 (Value of the Game)**.** A zero-sum game $G = (X, Y, u)$ possesses a **value** $V \in \mathbb{R}$ if:

$$V = \underline{v} = \overline{v}$$

That is, if the maximin equals the minimax.

Remark (Security Value). The minimax value $\overline{v}$ can be seen as a kind of "security value" for Player 2 (in the sense that Player 2 is sure not to have to pay more than a value very close to $\overline{v}$).

Proposition 6.6 (Fundamental Inequality)**.** *For any function u, we always have:*

$$\underline{v} \leq \overline{v}$$

The converse is false in general (unless there is a saddle point).

6.4 Topological Tools (Prerequisites)

Sion's theorem requires precise notions of continuity and convexity.

Definition 6.7 (Semi-continuity)**.** Let $f : X \to \mathbb{R}$.

- f is **upper semi-continuous (u.s.c.)** if for all λ, the upper level set $\{x : f(x) \geq \lambda\}$ is **closed**.
- f is **lower semi-continuous (l.s.c.)** if for all λ, the lower level set $\{x : f(x) \leq \lambda\}$ is **closed**.

Definition 6.8 (Quasi-convexity)**.** f is **quasi-convex** if its lower level sets $\{x : f(x) \leq \lambda\}$ are **convex**. (The opposite of a quasi-concave function).

6.4.1 The Intersection Lemma

This lemma is crucial for Sion's proof:

Lemma 6.9 (Separation (Preliminary))**.** *Let A and B be two non-empty, disjoint, compact convex sets. Let H be the hyperplane strictly separating them (Hahn-Banach Theorem). If a convex set C has a non-empty intersection with A and with B, then $C \cap H \neq \emptyset$.*

Lemma 6.10 (Intersection)**.** *Let $C_1, \ldots, C_n$ be non-empty compact convex sets. If $\cup_{i=1}^n C_i$ is convex and if for all $j \in \{1, \ldots, n\}$, $\cap_{i \neq j} C_i \neq \emptyset$, then $\cap_{i=1}^n C_i \neq \emptyset$.*

Proof (by induction on n).

- **Initialization ($n = 2$):** If $C_1 \cup C_2$ is convex with $C_1 \cap C_2 = \emptyset$, then by Hahn-Banach there exists a hyperplane H strictly separating C_1 and C_2. Since $C_1 \cup C_2$ is convex and intersects both sides of H, it must intersect H. However, H meets neither C_1 nor C_2, contradiction. Thus $C_1 \cap C_2 \neq \emptyset$.
- **Heridity:** Assume the result is true for $n-1$. (The rest of the proof constructs intersections of type $C_i \cap C_n$).

□

6.4.2 Von Neumann Matrix Notation

For a finite game defined by a matrix A, we denote the value:

$$val(A) = \max_{x \in \Delta(I)} \min_{y \in \Delta(J)} xAy = \min_{y \in \Delta(J)} \max_{x \in \Delta(I)} xAy$$

where x is a row vector and y a column vector.

6.5 Von Neumann's Minimax Theorem

This theorem establishes a deep connection between Nash equilibrium and maximin/minimax values in zero-sum games.

Theorem 6.11 (Von Neumann Minimax). *For a two-player zero-sum game, the following two statements are equivalent:*

1. *The game possesses a* ***Nash equilibrium*** (x^*, y^*).
2. *The game has a* ***value****, i.e.,* $\underline{v} = \overline{v} = V$*, and each player has an optimal strategy.*

Consequence: *If these conditions are met, the equilibrium payoff is* $(V, -V)$.

6.6 ϵ-Nash Equilibrium and Value

Definition 6.12 (ϵ-equilibrium). A profile (x, y) is an ϵ-equilibrium if no one can gain more than ϵ by deviating:

$$u(d_x, y) - \epsilon \leq u(x, y) \quad \text{and} \quad -u(x, d_y) - \epsilon \leq -u(x, y)$$

Proposition 6.13. *A game has a value if and only if for all* $\epsilon > 0$*, there exists an* ϵ*-equilibrium.*

6.7 Detailed Study: Sion's Minimax Theorem

6.7.1 Hypotheses and Definitions

Let (X, Y, u) be a game where:

- X is a compact and convex set.
- Y is a convex set.
- u is quasi-concave and upper semi-continuous (u.s.c.) in x.
- u is quasi-convex and lower semi-continuous (l.s.c.) in y.

6.7.2 Part 1: Constructing the Contradiction

Question 1 Explain why there exists a real v such that:

$$\sup_{x \in X} \inf_{y \in Y} u(x, y) < v < \inf_{y \in Y} \sup_{x \in X} u(x, y)$$

Solution: We assume by contradiction that the game has no value, i.e., $\underline{v} < \overline{v}$. By the density property of reals, we can choose a v between these two bounds. If the values are finite, we can take $v = \frac{\underline{v} + \overline{v}}{2}$. If $\overline{v} = +\infty$, we can take $v = \underline{v} + 1$.

Question 2 Explain why (1) $\forall x \in X, \exists y \in Y : u(x, y) < v$ and (2) $\forall y \in Y, \exists x \in X : u(x, y) > v$. **Solution:** (1) Since $\sup_x(\inf_y u(x, y)) < v$, then for any x, the minimal value $\inf_y u(x, y)$ is strictly less than v. There must exist a y (dependent on x) such that $u(x, y) < v$. (2) Similarly, $\inf_y(\sup_x u(x, y)) > v$ implies that for all y, P1's best possible payoff is above v, so there exists an x such that $u(x, y) > v$.

Question 3 For each $y \in Y$ and $\epsilon > 0$, let $X_y^\epsilon = \{x \in X : u(x, y) < v - \epsilon\}$. Prove that it is an open set. **Solution:** The complement of this set in X is $\{x \in X : u(x, y) \geq v - \epsilon\}$. Since the map $x \mapsto u(x, y)$ is upper semi-continuous (u.s.c.), its upper level sets are closed by definition. The complement X_y^ϵ is therefore open.

Question 4 Prove that for all $x \in X$, there exists $y \in Y$ and $\epsilon > 0$ such that $u(x, y) < v - \epsilon$. **Solution:** From Question 2, for all x, there exists y such that $u(x, y) < v$. Keeping $\epsilon = \frac{v-u(x,y)}{2} > 0$, we indeed get $u(x, y) = v - 2\epsilon < v - \epsilon$.

Question 5 Prove that $X \subset \bigcup_{y \in Y, \epsilon > 0} X_y^\epsilon$. **Solution:** Since every point x of X belongs to at least one set X_y^ϵ (from Question 4), the union of all these open sets necessarily covers X.

Question 6 Prove that there exists a finite set $Y_0 \subset Y$ and $\epsilon > 0$ such that $X \subset \bigcup_{y \in Y_0} X_y^\epsilon$. **Solution:** The set X is compact. By the Borel-Lebesgue property, from any open cover, we can extract a finite subcover. Thus points $y_1, \ldots, y_n$ and associated ϵ_i exist. Taking $\epsilon = \min(\epsilon_1, \ldots, \epsilon_n)$, we maintain the inclusion since $X_{y_i}^{\epsilon_i} \subset X_{y_i}^\epsilon$ for $\epsilon \leq \epsilon_i$.

Question 7 Prove that $Y' := \mathrm{conv}(Y_0)$ is compact and convex. **Solution:** Y' is the convex hull of a finite number of points. It is convex by definition. As we work in a finite-dimensional space (the span of Y_0), it is bounded. For compactness, any sequence in Y' has its mixture coefficients in the simplex (which is compact), so by Bolzano-Weierstrass, we can extract a convergent subsequence whose limit is still in Y'.

Question 8 Prove that: $\sup_{x \in X} \inf_{y \in Y_0} u(x, y) < v < \inf_{y \in Y'} \sup_{x \in X} u(x, y)$. **Solution:** (Left) Since X is covered by X_y^ϵ for $y \in Y_0$, for all x, there is a $y \in Y_0$ such that $u(x, y) < v - \epsilon$. The infimum over Y_0 is thus always less than v. (Right) Since $Y' \subset Y$, the infimum over Y' is greater than or equal to the infimum over Y, which itself is strictly greater than v by the hypothesis of Question 1.

6.7.3 Part 2: Symmetrization

Question 11 Prove that there exists a finite set $X_0 \subset X$ such that $Y' \subset \bigcup_{x \in X_0} Y_x^\epsilon$. **Solution:** We apply the same reasoning as before: Y' is compact and u is l.s.c. in y, so the sets $Y_x^\epsilon = \{y \in Y' : u(x, y) > v + \epsilon\}$ are open. By compactness of Y', we extract a finite subcover X_0.

Question 13 Prove that $\sup_{x \in X'} \inf_{y \in Y'} u(x, y) < v < \inf_{y \in Y'} \sup_{x \in X_0} u(x, y)$ (where $X' = \mathrm{conv}(X_0)$). **Solution:** This follows from the monotonicity of sup and inf over compact subsets X' and Y'.

Question 16 Justify the existence of minimal X_0 and Y_0 satisfying these inequalities. **Solution:** If an element of X_0 or Y_0 is not necessary to maintain the inequality, we can simply remove it from the finite set until the set is minimal for inclusion.

6.7.4 Part 3: Intersection Lemma and Contradiction

Question 17 For $x \in X_0$, let $A_x = \{y \in Y' : u(x, y) \leq v\}$. Prove that A_x is compact and convex. **Solution:**

- **Compact:** u is l.s.c. in y, so A_x is closed. Being contained in the compact Y', it is compact.
- **Convex:** u is quasi-convex in y, so its lower level sets are convex by definition.

Question 18 Prove that $\bigcap_{x \in X_0} A_x = \emptyset$. **Solution:** If the intersection contained a point y^*, we would have $u(x, y^*) \leq v$ for all $x \in X_0$. This would imply $\sup_{x \in X_0} u(x, y^*) \leq v$, which contradicts the right inequality of Question 13: $\inf_{y \in Y'} \sup_{x \in X_0} u(x, y) > v$.

Question 20 Prove that $\bigcup_{x \in X_0} A_x$ is not convex and is not equal to Y'. **Solution:** By the Intersection Lemma, if a family of compact convex sets A_x has an empty total intersection but all its subfamilies have non-empty intersection (by minimality of X_0), then their union is not convex. In particular, it cannot be equal to the convex set Y'.

Question 21 & 22 Prove the existence of $y_0 \in Y'$ such that $u(x, y_0) > v$ for all $x \in X'$. **Solution:** Since the union of A_x does not cover Y', there exists $y_0 \in Y' \setminus \bigcup A_x$. Thus y_0 belongs to no A_x, hence $\forall x \in X_0, u(x, y_0) > v$. By quasi-concavity of u in x, this property holds for any convex combination $x \in X' = \text{conv}(X_0)$.

Question 23 Finish the proof. **Solution:** By a symmetric argument, there exists $x_0 \in X'$ such that for all $y \in Y'$, $u(x_0, y) < v$. Evaluating at point (x_0, y_0), we get $u(x_0, y_0) > v$ and $u(x_0, y_0) < v$. This is a contradiction. Thus $\underline{v} = \overline{v}$.

Question 25 Prove the existence of an optimal strategy. **Solution:** The map $f(x) = \inf_{y \in Y} u(x, y)$ is the infimum of a family of u.s.c. functions, so it is u.s.c. By the Weierstrass theorem, a u.s.c. function on a compact set (X) attains its maximum. Thus there exists an $\overline{x}$ such that $f(\overline{x}) = \sup_x f(x)$, which is the optimal strategy.

Exercise 6.14 (Sion-Wolfe Counter-Example)**.** Consider the game on $S_1 = S_2 = [0, 1]$ with:

$$u_1(x, y) = \begin{cases} -1 & \text{if } x < y < x + 1/2 \\ 0 & \text{if } x = y \text{ or } y = x + 1/2 \\ 1 & \text{otherwise} \end{cases}$$

Show that this game has no value (Min-Max differs from Max-Min) because Sion's regularity conditions (semi-continuity) are not satisfied.

Solution:

- **Maximin Calculation ($\underline{v}$):** For fixed x, Player 2 can choose $y = x + \epsilon$ (with small ϵ). Then $x < y < x + 1/2$, so $u = -1$. Thus $\inf_y u(x, y) = -1$. And $\sup_x(-1) = -1$. So $\underline{v} = -1$.
- **Minimax Calculation ($\overline{v}$):** For fixed y, Player 1 can choose $x = y$ or very close. If P1 chooses $x = y$, $u = 0$. But P1 can choose x such that y is not in $]x, x + 1/2[$. For example x such that $x > y$. Then $u = 1$. Thus $\sup_x u(x, y) = 1$. And $\inf_y(1) = 1$. So $\overline{v} = 1$.
- **Conclusion:** $-1 \neq 1$. The game has no value. This comes from the crude discontinuity of payoffs.

6.8 Complementary Exercises

6.8.1 4x4 Matrix Exercise

Calculate $\underline{v}$ and $\overline{v}$ for:

$$M = \begin{pmatrix} -1 & 2 & -1 & 0 \\ 1 & 3 & 3 & 5 \\ -9 & -2 & 8 & 2 \\ -1 & 4 & -4 & 5 \end{pmatrix}$$

Method:

- Maximin (Row): Min of each row $\rightarrow (-1, 1, -9, -4)$. The max is 1 (Row 2). So $\underline{v} = 1$.
- Minimax (Column): Max of each column $\rightarrow (1, 4, 8, 5)$. The min is 1 (Column 1). So $\overline{v} = 1$.
- Conclusion: Saddle point $(R2, C1)$, value $v = 1$.

6.8.2 "No Value" Exercise (Course)

Consider the zero-sum game given by the following matrix (P1 actions: i, P2 actions: j):

$$A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 5 \\ 9 & -2 & 8 & 2 \\ 1 & 4 & -4 & 5 \end{pmatrix}$$

Payoff Analysis: For example, if $i = 1$ and $j = 2$ (P1 plays 1, P2 plays 2), then P1 wins 2 and P2 wins -2.

Pure Value Calculation:

- **Maximin ($\underline{V}$)**:
  - Row 1: min = -1
  - Row 2: min = 0
  - Row 3: min = -2
  - Row 4: min = -4

  The maximum of these minimums is $\underline{V} = 0$.
- **Minimax ($\overline{V}$)**:
  - Col 1: max = 9
  - Col 2: max = 4
  - Col 3: max = 8
  - Col 4: max = 5

  The minimum of these maximums is $\overline{V} = 4$.

Conclusion:

$$\underline{V} = 0 < 4 = \overline{V}$$

The game has **no value** (in pure strategies). Thus there is no Nash equilibrium and no consensus possible.

6.8.3 2x2 Game without Pure Value (Course Example)

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

1. **Pure Values:**
   - Maximin (J1): min row 1 = 1, min row 2 = 0. Max = 1. So $\underline{v} = 1$.
   - Minimax (J2): max col 1 = 2, max col 2 = 2. Min = 2. So $\overline{v} = 2$.

   Since $\underline{v} < \overline{v}$, there is no saddle point (no consensus).
2. **Mixed Strategies (Indifference Method):** Let p be the probability P1 plays Top, and q be the probability P2 plays Left.
   - **P1 Indifference:** P1 must earn the same if P2 plays L or R (for P2 to be willing to mix). *Note: Course notes might invert logic, but theoretically:* P1 makes P2 indifferent:

$$2p + 0(1-p) = 1p + 2(1-p)$$

$$2p = p + 2 - 2p \implies 3p = 2 \implies p = 2/3$$

- **P2 Indifference:** P2 makes P1 indifferent:

$$2q + 1(1-q) = 0q + 2(1-q)$$

$$q + 1 = 2 - 2q \implies 3q = 1 \implies q = 1/3$$

Nash Equilibrium: $((2/3, 1/3), (1/3, 2/3))$. **Value of the game:** $v = 3(1/3)+1(2/3)\ldots$ no, let's calculate P1's payoff: $u_1 = 2(1/3)+1(2/3) = 4/3$ (if P1 plays T against q). Check: $0(1/3) + 2(2/3) = 4/3$. OK.

Exercise 6.15 (Link between Value and ϵ-Nash)**.** Prove that a zero-sum game has a value V if and only if, for all $\epsilon > 0$, there exists a strategy profile $(x^*_\epsilon, y^*_\epsilon)$ such that:

$$\underline{v}(x^*_\epsilon) \geq V - \epsilon \quad \text{and} \quad \overline{v}(y^*_\epsilon) \leq V + \epsilon$$

Solution:

- $\Rightarrow$ If V exists, $\underline{v} = \overline{v} = V$. It suffices to take x^*_ϵ an ϵ-optimal strategy for maximin, and y^*_ϵ for minimax.
- $\Leftarrow$ We always have $\underline{v} \leq \overline{v}$. The hypothesis gives: $V - \epsilon \leq \underline{v}(x^*_\epsilon) \leq \underline{v} \leq \overline{v} \leq \overline{v}(y^*_\epsilon) \leq V + \epsilon$. Letting $\epsilon \to 0$, we get $V \leq \underline{v} \leq \overline{v} \leq V$, so equality holds everywhere.

Chapter 7

Extensive Form Games

7.1 Formalization: Tree and Play

An extensive form game models sequential interactions. It is represented by a **game tree** T.

7.1.1 The notion of "Play"

Although intuitive, the distinction between strategy and play is fundamental.

Definition 7.1 (Play)**.** A **play** is a complete path starting from the root of the tree and ending at a leaf (terminal node).

Proposition 7.2. *Given a pure strategy profile $x = (x_1, \ldots, x_n)$, this generates a **unique play** $\pi(x)$. However, several different strategy profiles can generate the same play (if their differences are only at unreached nodes).*

7.2 Game Tree Representation

A game tree contains:

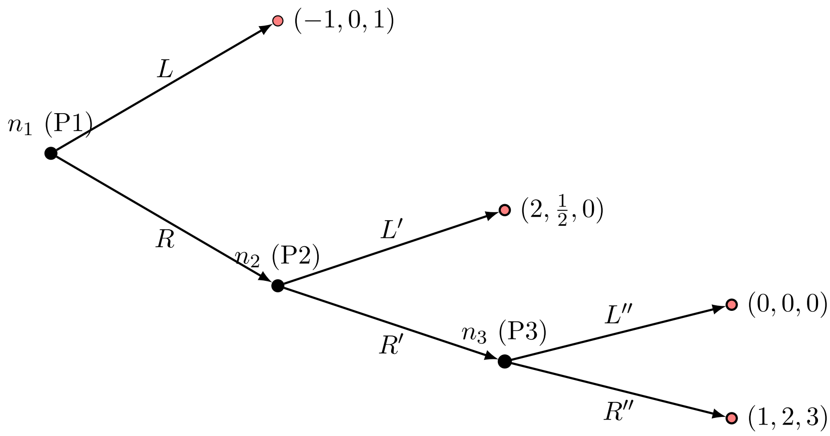
- **Decision nodes:** Identify which player is to move.
- **Branches:** Possible actions at each node.
- **Leaves:** Final outcomes and payoffs.

7.2.1 Nice Extensive Form Games

Definition 7.3. A "Nice Extensive Form Game" is an extensive form game satisfying:

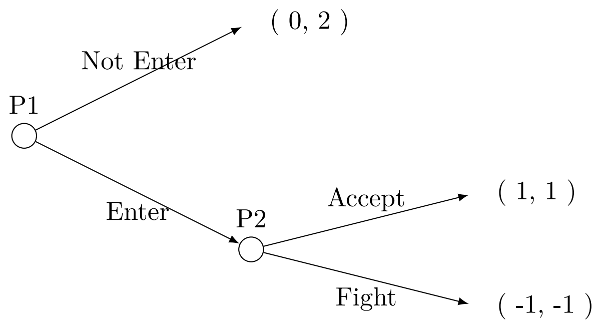
1. A **finite** tree (finite horizon, finite nodes, finite edges).
2. At each node (except terminal ones), there is a **unique** designated player.
3. At terminal nodes, there are real payoff vectors (one for each player).

Example 7.4 (3-Player Game - Course Diagram)**.**



Legend: Players in black $(1, 2, 3)$, terminal nodes in red. The story: Player 1 chooses L or R. If they choose L, the game ends. If they choose R, it's Player 2's turn, etc.

Example 7.5 (Market Entry Game)**.** Firm A (Entrant) decides to Enter (E) or Not (N). If it enters, Firm B (Incumbent) decides to Fight (B) or Accept (A).



7.3 Information and Strategies

7.3.1 Perfect vs Imperfect Information

- **Perfect Information:** Each player knows the complete history of the game (all nodes are singletons).
- **Imperfect Information:** Some nodes are grouped into **information sets**. The player knows they are in the set, but not at which specific node.

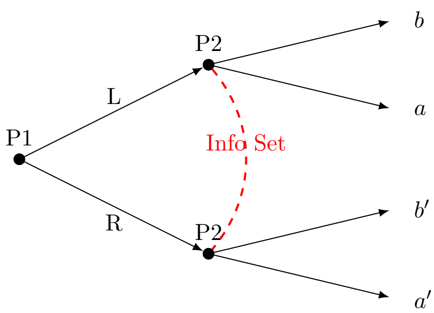


Figure 7.1: Information Set: P2 does not know if P1 played L or R.

7.3.2 Condensed Strategy Notation

A major distinction of the course:

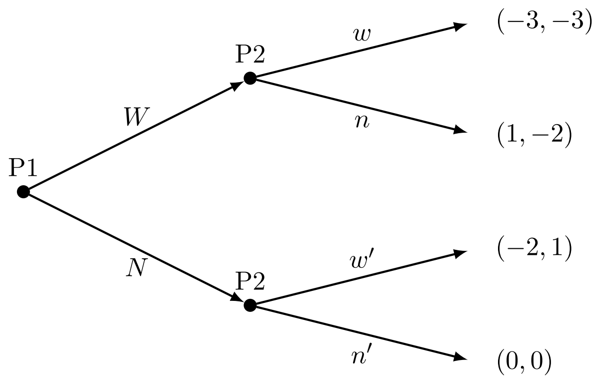
Definition 7.6 (Strategy)**.** A **strategy** is a **complete** and **exhaustive** plan of action: it specifies an action for *every* information set where the player is called to play, **even those that will never be reached** if the strategy is followed.

7.3.3 Normal Form associated with Extensive Form

Definition 7.7. To any extensive form game Γ, we can associate a normal form game (matrix) defined by:

- The same players.
- Strategies defined as complete action plans (e.g., $W'N'$).
- Payoffs determined by the unique play generated by each strategy profile.

Example 7.8 ("Nuclear Weapon" Game - Course Example)**. 1. Game Tree (Γ):** Player 1 (P1) chooses Nuclear Weapon (W) or Not (N). Player 2 (P2) reacts.



Note: The notations w, n (left) and w', n' (right) for P2 allow constructing their strategies.
2. Strategies:

- P1: $\{W, N\}$
- P2: $\{ww', wn', nw', nn'\}$ (Often denoted $W'W', W'N'$ etc. in notes). Example: wn' means "I play w if P1 plays W, and I play n' if P1 plays N".

3. Payoff Matrix (Normal Form): Each cell corresponds to the payoff of the reached leaf.

$P1 \setminus P2$	ww'	wn'	nw'	nn'
W	$(-3,-3)$	$(-3,-3)$	$(1,-2)$	$(1,-2)$
N	$(-2,1)$	$(0,0)$	$(-2,1)$	$(0,0)$

Explanation: If P1 plays W and P2 plays wn', the game goes to W then to w. Payoff $(-3,-3)$.

7.4 Backward Induction and Credibility

7.4.1 Backward Induction

To solve a finite game with perfect information, we reason backwards:

1. Focus on the last decision nodes.
2. The player chooses the action maximizing their payoff.

3. Replace the subgame with these payoffs (we "prune" the tree).

4. Move up to the root.

Theorem 7.9. *The backward induction algorithm determines a* ***Subgame Perfect Nash Equilibrium (SPE)****.*

Definition 7.10 (Subgame Perfect Equilibrium - SPE)**.** A strategy profile is a **Subgame Perfect Equilibrium (SPE)** if it induces a Nash equilibrium in **every subgame** of the tree (including the whole game).

Example 7.11 (Exam: SPE vs Nash)**.** Consider the following game: P1 chooses A or D (payoff 2,2). If A, P2 chooses l or r. If l, P1 chooses $L(4,2)$ or $R(1,0)$. If r, P1 chooses $L'(0,3)$ or $R'(3,1)$.

- **Normal Form:** P1 has 4 strategies $\{AL, AR, DL, DR\}$. P2 has 2 $\{l, r\}$.
- **Nash Equilibria:** (AL, r) (Payoff $0, 3$), (DL, l) (Payoff $2, 2$), (DR, l) (Payoff $2, 2$).
- **Subgame Analysis (Induction):**
  1. Subgame after (A, l): P1 chooses L $(4 > 1)$.
  2. Subgame after (A, r): P1 chooses R' $(3 > 0)$.
  3. P2 anticipates: if they play l, P1 plays L (P2 payoff $= 2$). If they play r, P1 plays R' (P2 payoff $= 1$). P2 chooses l.
  4. Root: P1 anticipates that A leads to L (payoff 4). D leads to 2. P1 chooses A.
- **Conclusion:** The unique SPE is the path (A, l, L) given by the complete strategies **(P1:** A, L, R'**; P2:** l**)**.
- **Analysis:** The Nash equilibrium (DL, l) is not an SPE because it relies on the (implicit) threat that P1 would play poorly in the subgame (A, r) (they would not play their best response).

7.5 Practical Case: The Centipede Game

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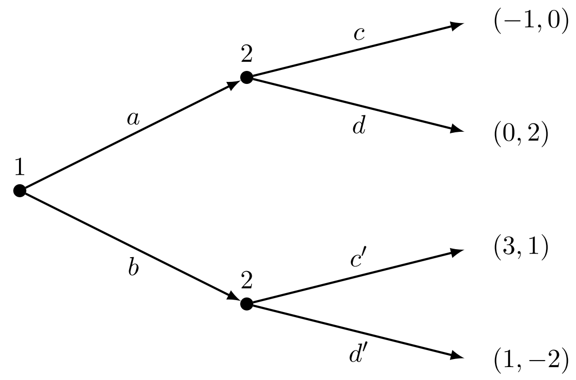
7.6 Exercises

7.6.1 Exercise "Tree and Matrix" (Handwritten Notes)

Consider the following game: P1 chooses a or b. P2 then chooses (c, d if a; c', d' if b).



Question: Calculate the Nash Equilibria.
Answer: We construct the 2×4 matrix. P2 Strategies: cc', cd', dc', dd'.

$1 \setminus 2$	cc'	cd'	dc'	dd'
a	$(-1, 0)$	$(-1, 0)$	$(\mathbf{0}, \mathbf{2})$	$(\mathbf{0}, \mathbf{2})$
b	$(\mathbf{3}, \mathbf{1})$	$(1, -2)$	$(\mathbf{3}, \mathbf{1})$	$(1, -2)$

- If P1 plays a: P2 prefers d (payoff 2) $\to$ columns dc', dd'. P1 responds a or b? Compare 0 and 3. P1 prefers b. So no pure Nash on row a.
- If P1 plays b: P2 prefers c' (payoff 1) $\to$ columns cc', dc'. P1 responds b (3 vs -1 or 0).
- **Nash Equilibria:** (b, cc') and (b, dc').
- Equilibrium payoff: $(3, 1)$.

7.7 Backward Induction and Credibility

7.7.1 Backward Induction

To solve a finite game with perfect information, we reason backwards:

1. Focus on the last decision nodes.
2. The player chooses the action maximizing their payoff.
3. Replace the subgame with these payoffs (we "prune" the tree).
4. Move up to the root.

Theorem 7.12. *The backward induction algorithm determines a **Subgame Perfect Nash Equilibrium (SPE)**.*

7.7.2 Non-Credible Threats

Every SPE is a Nash equilibrium, but the converse is false.

- A Nash equilibrium can rely on a "non-credible threat": the player promises a severe punishment (e.g., blowing themselves up) at a node that is not reached in equilibrium.
- Since the node is never reached, the threat costs nothing ("talk is cheap").
- The concept of SPE (and backward induction) eliminates these equilibria by demanding rationality **everywhere**, even off the equilibrium path.

7.8 Practical Case: The Centipede Game

Description: Two players take turns. At each round, a player can "Stop" (taking the large pile, leaving little to the other), or "Pass" (to increase the global pot). The game has a fixed end (e.g., 100 rounds).

Paradox of Backward Induction:

- At the last round (100), the deciding player has an incentive to Stop to take the payoff.
- Anticipating this, at round 99, the other player has an incentive to Stop to avoid getting nothing at round 100.
- By induction, backward induction predicts that **the first player stops at the very first round**!

Conclusion: Although rational in the SPE sense, this result is sub-optimal (Pareto-inferior to cooperation) and rarely observed in practice.

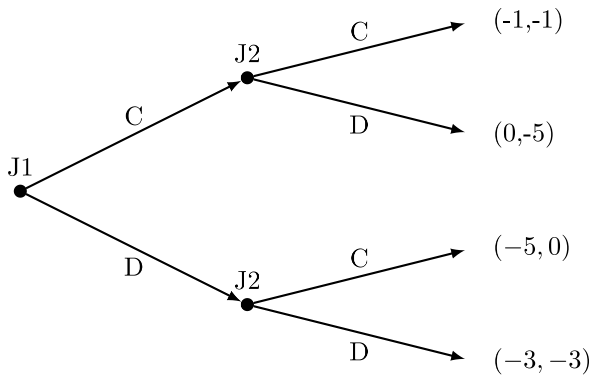
7.9 Exercises

Exercise 7.13 (Extensive Form Games)**.**

- Represent the Sequential Prisoner's Dilemma.
- Verify that in the entry game, the threat "Fight" is not credible if the cost of fighting is too high.

Solution:

1. **Sequential PD:** J1 plays (C or D). Then J2 plays (C or D) knowing J1's choice.



Backward Induction:

- Top (if J1 plays C): J2 chooses D ($0 > -1$).
- Bottom (if J1 plays D): J2 chooses D ($-3 > -5$).
- J1 anticipates that C leads to $(0, -5)$ (for J2 it's payoff 0, for J1 it's -5?) *Watch the payoffs (J1, J2).*
- Let's verify payoffs: If (C, C) $\to (-1, -1)$. If (C, D) $\to (-5, 0)$. J2 gets 0. If (D, C) $\to (0, -5)$. J2 gets -5. If (D, D) $\to (-3, -3)$.
- **J2 Node (Top, after C):** J2 compares C (-1) and D (0). Chooses D.
- **J2 Node (Bottom, after D):** J2 compares C (-5) and D (-3). Chooses D.

- **J1 Root:** Anticipates (C $\rightarrow$ J2 plays D $\rightarrow$ J1 Payoff = -5) and (D $\rightarrow$ J2 plays D $\rightarrow$ f J1 Payoff = -3).
- J1 compares -5 and -3. Chooses D.
- **Result:** (D, D).

2. **Entry Game:** If the Entrant enters, the Incumbent has the choice between "Fight" (Payoffs: -1, -1) or "Accept" (Payoffs: 1, 1). Since $1 > -1$, the rational Incumbent will **always** choose to Accept. Any strategy where they threaten to fight is non-credible (does not withstand backward induction). The Entrant knows this, so they Enter (since $1 > 0$).